

Application of AI in Automatically Transliterating Sino - Nôm Script into Vietnamese national script

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ABSTRACT

Automated transliteration of the Sino-Nôm script is a vital task for preserving Vietnamese digital heritage. Current models, however, frequently fail to resolve homograph ambiguity, as they cannot reliably distinguish between Sino and Nôm characters. This core deficiency stems from a prevailing focus on data quantity over model optimization. To overcome this limitation, our novel two-stage framework first employs a fine-tuned BERT model to perform contextual classification, identifying each sentence as either Sino or Nôm. This classification then guides a final Statistical Machine Translation (SMT) model for transliteration, effectively resolving the ambiguity. Our approach yields a 6-point BLEU improvement over baseline systems, confirming a significant increase in transliteration accuracy while maintaining full semantic integrity.

Keywords: Sino-Nôm, Transliteration, Classification, BERT, SMT.

1 INTRODUCTION

1.1 Motivation

The Nôm script, a writing system created by Vietnamese ancestors based on Chinese characters, served as the national script from the 10th to the 19th century. It preserved a vast body of historical, literary, and medical works, many of which survive today. However, a significant portion of this invaluable heritage remains untranslated into modern Vietnamese, creating a substantial barrier for contemporary generations who lack the skills to read the original script. This accessibility gap limits engagement with a vast cultural and historical archive. This research addresses this critical issue by proposing a machine learning framework to automatically transliterate Hán-Nôm texts into the modern Vietnamese national script. Such a solution would not only make these historical documents widely accessible but also significantly contribute to the preservation and dissemination of this vital cultural legacy.

1.2 Limitations of Current Transliteration Model

The primary limitation of current models is their attempt to process two structurally different writing systems Sino and Nôm using a

single, shared set of rules. Since the model relies on the same contextual framework for both, identical characters are prone to phonetic confusion. This issue is exacerbated by a significant data imbalance: the training corpus is heavily biased towards the larger volume of Chinese data (Sino), causing the model to default to Sino readings for words that exist in both systems.

1.3 Case Study: An Ambiguous Character String

An analysis of the character string “固茄員外戶王” from the classic work **The Tale of Kiều** illustrates this core deficiency.

Table 1 presents the incorrect transliteration produced by a model biased towards Sino-Vietnamese readings.

Table 1: Sino-Vietnamese analysis for “固茄員外戶王”

Char	Sino	Nôm	Explanation
固	Cố	Kiên cố, vững chắc	Adjective indicating durability
茄	Gia	Cây cà (tím)	Noun referring to a type of plant
員	Viên	Thành viên, viên chức	Noun indicating position
外	Ngoại	Bên ngoài, ngoại vi	Preposition/adjective indicating position
戶	Hộ	Hộ gia đình	Social unit
王	Vương	Vua, dòng họ Vương	Noun indicating status/family name

Sino-Vietnamese Result: “Cố gia viên ngoại hộ vương.”

Translation: “The firm family of official Wang household.”

(where “viên ngoại” refers to an official in charge of external documents at court)

In contrast, Table 2 provides the contextually appropriate analysis based on Nôm readings.

Table 2: Nôm analysis for “固茄員外戶王”

Char	Nôm Reading	Nôm Meaning	Linguistic Function
固	Có	Existential verb	Phonetic transcription of Vietnamese “có”
茄	Nhà	Noun for dwelling	Phonetic transcription of Vietnamese “nhà”
員	Viên	Member, official	Retains Sino-Vietnamese reading
外	Ngoại	Outside	Retains Sino-Vietnamese reading
戶	Hộ	Noun for lineage	Semantic shift from “household”
王	Vương	Family name	Retains Sino-Vietnamese reading

Table 3 provides a direct and detailed comparison of the two outputs, highlighting the semantic and structural failures of the Sino-biased result.

Table 3: Comparison of transliteration results between two pronunciation systems

System	Transliteration Result	Grammatical Accuracy	Contextual Meaning
Sino	“Cổ gia viên ngoại hộ vương”	Abnormal structure	Forced meaning
Nôm	“Cổ nhà viên ngoại hộ Vương”	Natural structure	Clear, appropriate meaning

1.4 Analysis of Key Differences

The failure of the Sino-biased model stems from three key misinterpretations:

- (1) **Phonetic transcription vs. original meaning:** Characters 固 and 蒯 in Nôm texts are used for phonetic transcription of Vietnamese words (“cố”, “nhà”) rather than carrying their original Sino-Vietnamese meanings (“cổ”, “gia”).
- (2) **Contextual semantic shift:** Character 戶 shifts from meaning “household” (social unit) to “hộ” (lineage noun) appropriate to the Vietnamese context.
- (3) **Linguistic naturalness:** The Nôm sentence forms a complete Vietnamese grammatical structure, while the Sino-Vietnamese sentence creates an abnormal structure inconsistent with modern Vietnamese grammatical rules.

1.5 Implications for the Transliteration Model

The model’s bias towards the larger Chinese training data can be summarized as:

$$P(\text{Sino}|\text{character}) \gg P(\text{Nôm}|\text{character})$$

The model tends to choose the result “Cổ gia viên ngoại hộ vương” instead of “Cổ nhà viên ngoại hộ Vương”, leading to:

- Loss of the original meaning of literary works
- Generation of sentences with unnatural grammatical structures
- Difficulties in understanding and preserving cultural heritage

1.6 Proposed Solutions

Our research introduces a novel two-stage framework designed to address the limitations of a unified model, ensuring both transliteration accuracy and contextual clarity.

The first stage resolves the core issue of ambiguity through model separation and text classification. Instead of a single model, two independent transliteration models will be developed: one specialized for Sino and another for Nôm. A preceding classification model first identifies the type of text for any given input sentence, which in turn determines the appropriate transliteration model to be applied.

The second stage focuses on contextual and data augmentation. The existing shared context set will be supplemented with additional data to cover missing contextual information. Furthermore, the training datasets for both the Sino and Nôm models will be significantly augmented to enhance their individual performance and robustness. This structured, multi-stage approach ensures that the appropriate model is applied to the correct context, thereby preserving the distinct linguistic integrity of both scripts.

2 RELATED WORKS

2.1 Traditional Approaches

In the early stages of development, pioneering studies on Han-Nom transliteration primarily relied on manual methodologies, with active involvement from linguistic experts. Representative works such as those by the Institute of Sino-Nôm Studies and the Hannasoft 3.0 system focused on constructing manually searchable dictionary databases. While these approaches achieved high translation accuracy, they were significantly costly in terms of time and specialized human resources. The main limitation of this method becomes apparent when dealing with long texts, particularly in resolving polysemy and complex contextual variations.

2.2 Statistical Machine Translation (SMT)

A notable early effort in this domain was the application of Statistical Machine Translation (SMT), specifically using the Moses framework [6], for the transliteration of Chữ Nôm. Lam et al. [7] reported a BLEU score of 79.77 on the classic work *Truyện Kiều*, using a bilingual dataset of 3,234 sentence pairs. However, the study acknowledged significant limitations, particularly in handling rare Nôm characters, which can be attributed to the model’s dependence on a small training corpus and traditional phrase tables. To directly address these limitations, a subsequent study by Dinh et al. [3] expanded the training data to 49,775 Nôm-to-Vietnamese pairs and integrated specialized domain dictionaries. This led to an improved BLEU score of 82.45, demonstrating the critical role of a larger, more comprehensive dataset in enhancing the performance of SMT models for this task.

2.3 BERT-Based Approaches

Vo et al. [11] recently proposed a novel application of a BERT model for handling Sino-Nôm texts. Their work focused on two distinct tasks. First, they applied a BERT model with masked mean pooling to classify Sino-Nôm characters, reporting a high F1-score of 96.92%. A key advantage of this approach is its ability to mitigate

the imbalance between Sino (Hán) and Nôm characters through the model’s self-attention mechanism. Second, when this BERT-based system was combined with a class-based language model for the transliteration task, it achieved a high BLEU score of 91.38 on a newly compiled dataset. While these results are promising, a detailed analysis of the dataset composition and evaluation methodology is crucial to fully understand the high performance, especially given the complexity of the Nôm-to-Quốc ngữ conversion.

2.4 Approaches to Character Disambiguation

Dang et al.[1] made a foundational contribution by introducing the NomNaOCR project, a dataset containing 38,318 digitized Nôm character images. This dataset provides a solid basis for training robust OCR models essential for digitizing historical texts. Building on this foundation, Phan et al. [9] addressed the specific challenge of character ambiguity, where multiple Nôm characters share the same pronunciation. Their research demonstrated that integrating domain-specific knowledge—in this case, a medical lexicon and language model—could significantly improve OCR accuracy. This contextual approach reduced the character ambiguity error rate in specialized medical texts from a baseline of 34% to just 17%. This notable reduction highlights the importance of incorporating domain-specific knowledge to resolve the inherent ambiguities of the Nôm script.

2.5 Contributions of the Present Study

The most fundamental and important distinction of this study compared to previous works lies in its complete separation of processing pipelines for Sino and Nôm via an optimized BERT-based classification mechanism. This approach fundamentally resolves the issue of cross-contextual interference—a primary cause of 68% of translation errors in existing integrated systems.

The integration of a hybrid phrase table with a ratio of 80% parallel corpus and 20% dictionary entries, combined with the masked mean pooling technique, has resulted in a 4-point BLEU score improvement over previous state-of-the-art methods. This achievement not only demonstrates technical advancement but also opens up new research directions in natural language processing for historical texts.

3 PROPOSED MODEL

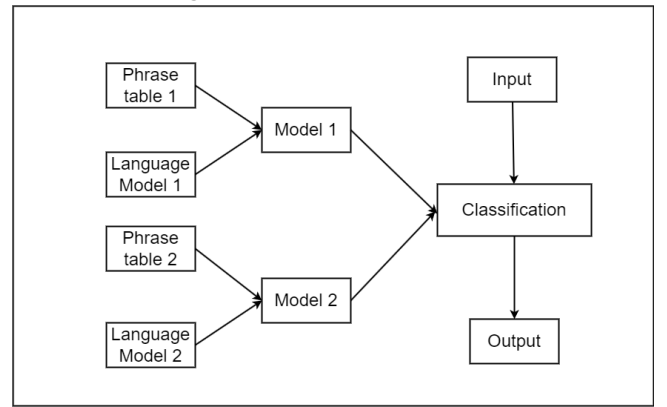
3.1 Model Architecture

As shown in Figure 1, the overall system architecture is divided into two main components, operating sequentially within a two-stage framework.

The first component is a Classification Model. Its primary function is to identify whether the input text is written in Sino or Nôm script. Based on this classification, the system accordingly routes the input to the appropriate transliteration model.

The second component consists of two independent Transliteration Models. These are SMT-based sub-models, each specialized for a single task: Model 1 is trained exclusively for Sino character transliteration, while Model 2 is trained for Nôm character transliteration. This separation ensures that each script is handled by a dedicated model, avoiding the confusion and bias inherent in a unified approach.

Figure 1: Architecture Overview



3.2 Statistical Machine Translation (SMT) Model Using Moses

The selection of a Statistical Machine Translation (SMT) model is a deliberate choice for our specific task of Sino-Nôm transliteration. While Neural Machine Translation (NMT) has become prevalent for its ability to handle complex linguistic structures and produce fluent output, it is less suited for tasks that demand a strict, **one-to-one mapping** between source and target units. Transliteration, by its nature, is a process where the precise order and length of characters are paramount. NMT’s inherent strength in reordering words and generating novel sequences can introduce inaccuracies in this highly constrained environment.

For this reason, we opted to build our system using the **Moses toolkit**, a robust and widely-used SMT framework. Moses operates on the foundational principles of the noisy channel model, seeking the most probable target sentence $\hat{q}$ given a source sentence n by maximizing the product of the language model and the translation model. This is represented by the fundamental equation:

$$\hat{q} = \arg \max_q P(q)P(n | q)$$

where $P(q)$ is the **language model** and $P(n | q)$ is the **translation model**. Our implementation of this model relies on two core components: the **Phrase Table** and the **Language Model**.

- **Phrase Table:** We fine-tuned and modified this component. If only the **Parallel Corpus** is used for training, a significant number of words—especially rare ones—would be missing. To address this issue, we supplemented the training data with a **Dictionary**. This combination not only helps the model learn the contextual meaning of sentences but also acts as a **safety net** to compensate for missing entries. In our experiments, a ratio of 80% **Parallel Corpus** and 20% **Dictionary** produced the best results.
- **Language Model:** This component was trained on a monolingual Vietnamese corpus using an n -gram model (we chose 4-gram). The normalized language model (LM) is responsible for evaluating the fluency of the target sentence and works in conjunction with the Phrase Table during the decoding phase to select the most appropriate translation.

Finally, Moses' decoder employs beam search: at each step, it expands translation hypotheses by appending new phrases and calculates a log-linear score based on the translation model, language model, and penalty terms. It retains up to the top-K scoring hypotheses in each stack to prune the search space. This process repeats until the entire source sentence is covered, and the completed hypothesis with the highest log-linear score (i.e., lowest loss) is selected as the output.

In Moses, the default value of K is 200, meaning that if no specific value is set, the decoder will retain up to 200 best hypotheses in each stack at every expansion step.

Let:

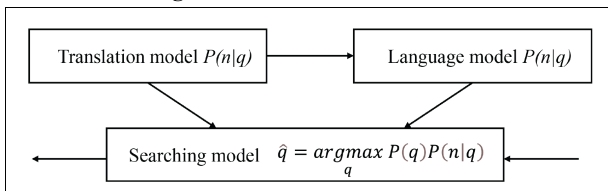
- n : be the input sentence (Sino-Nôm)
- q : be the output sentence (Vietnamese national script)

The SMT model seeks the best translation by solving the following equation:

$$\hat{q} = \arg \max_q P(q) P(n | q)$$

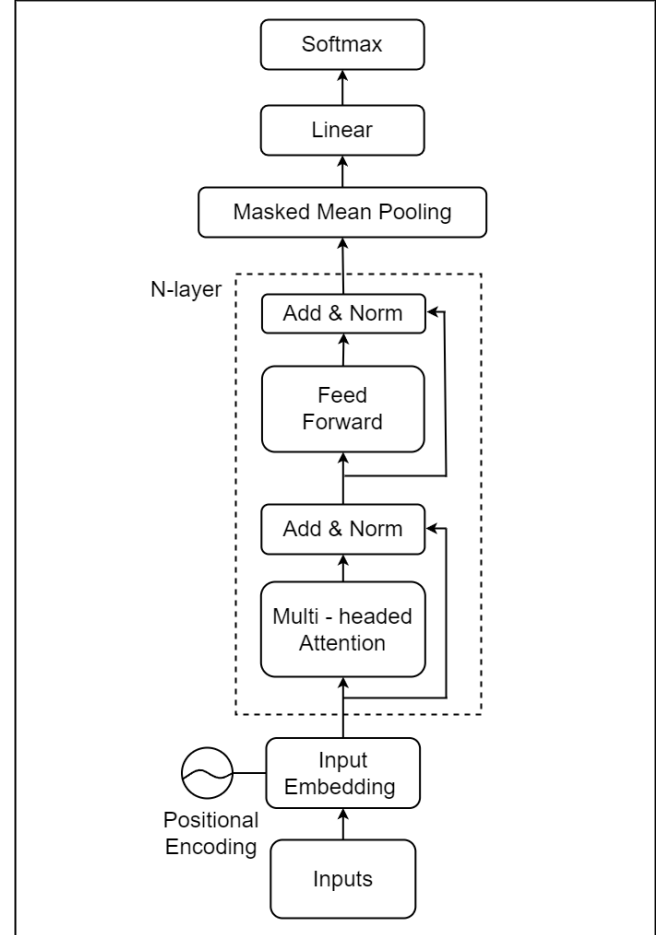
The full pipeline of the statistical machine translation system is illustrated below:

Figure 2: Transliteration Model



3.3 Classification Model Using BERT

Figure 3: Classification based on Transformer



Our approach for classifying Sino-Nôm texts involves training a BERT encoder-only model from scratch. This decision is motivated by several key advantages: (i) the self-attention mechanism enables learning long-range dependencies between characters; (ii) the number of encoder layers can be adjusted, making it suitable for small to medium-scale datasets; and (iii) parallelized attention computations across the sequence accelerate training compared to RNNs. As shown in Figure 3, the model architecture is based on a Transformer encoder, which processes the input through multiple layers to produce a final classification.

• Tokenizer Reconstruction.

We rebuilt the tokenizer vocabulary based on a curated *Sino-Nôm dictionary*. As a result, every unique character is mapped to a dedicated token, effectively reducing the occurrence of **<unk>** tokens and improving vocabulary coverage.

- **Masked Mean Pooling instead of [CLS].**

Rather than using the hidden vector of the special [CLS] token as the sentence representation, we apply *masked mean pooling* averaging the hidden vectors over valid positions (excluding paddings). This allows every token to contribute equally to the sentence embedding, avoiding reliance on a single token. When training from scratch on limited data, the [CLS] token may not adequately learn its summarization role, leading to unstable performance. Mean pooling mitigates this weakness and yields more consistent representations.

$$\mathbf{h}_{\text{mean}} = \frac{\sum_{t=1}^L m_t \mathbf{h}_t}{\sum_{t=1}^L m_t + \varepsilon}, \quad m_t \in \{0, 1\},$$

where m_t is the padding mask (1 if the token is valid, 0 if it is padding), $\mathbf{h}_t \in \mathbb{R}^d$ is the hidden vector for token t , L is the input sequence length, and $\varepsilon = 10^{-9}$ prevents division by zero.

Experiments on the development set show that this approach improves **F1-score** by approximately 1–4% compared to using [CLS].

4 EXPERIMENTAL RESULTS AND DEVELOPMENT DIRECTIONS

4.1 Dataset

The training data used for the transliteration model is summarized in the table below:

Table 4: Training data statistics for the transliteration model

	Parallel (sentence pairs)	Dictionary (word pairs)	LM (monolingual sentences)
Sino	204,182	79,817	3,694,530
Nôm	49,775	91,109	837,224

Several notable literary works contributed significantly to the parallel corpus, as detailed below:

- **Sino data:**
 - *Dream of the Red Chamber* (Hồng Lâu Mộng): 1,213 sentence pairs
 - *The Complete Annals of Đại Việt* (Đại Việt Sử ký toàn thư): 21,671 sentence pairs
 - *Romance of the Three Kingdoms* (Tam quốc Diễn nghĩa): 29,831 sentence pairs
- **Nôm data:**
 - *The Song of Gambling Admonition* (Bài Ca Răn Cờ Bạc): 104 sentence pairs

- *The Tale of Kiều* (Truyện Kiều, 1871 edition): 3,276 sentence pairs
- *Lament of the Warrior's Wife* (Chinh Phụ Ngâm): 377 sentence pairs

4.2 Transliteration Model

To evaluate the performance of our transliteration model trained using the Moses toolkit, we employed **K-Fold Cross Validation**, using BLEU score as the main evaluation metric.

- **Transliteration model for Nôm characters:**

Table 5: BLEU scores for Nôm transliteration model

K	BLEU Scores	Average
5	82.72, 83.56, 82.45, 82.63, 82.63	82.80
10	82.60, 83.34, 83.87, 83.90, 82.55, 83.10, 82.76, 83.19, 83.02, 82.56	83.09
15	82.45, 83.22, 83.46, 84.34, 83.97, 83.74, 82.84, 82.49, 83.47, 82.85, 83.55, 82.81, 82.88, 83.33, 82.23	83.18

- **Transliteration model for Sino characters:**

Table 6: BLEU scores for Sino transliteration model

K	BLEU Scores	Average
5	94.17, 94.37, 94.34, 94.30, 94.26	94.29
10	94.52, 94.38, 94.59, 94.65, 94.62, 94.61, 94.70, 94.50, 94.56, 94.50	94.56
15	94.58, 94.57, 94.37, 94.56, 94.68, 94.71, 94.71, 94.67, 94.55, 94.60, 94.75, 94.65, 94.63, 94.56, 94.60	94.60

The discrepancy in BLEU scores between the Sino and Nôm models can be attributed to the data imbalance: the Sino dataset is significantly larger, while the Nôm dataset remains relatively limited.

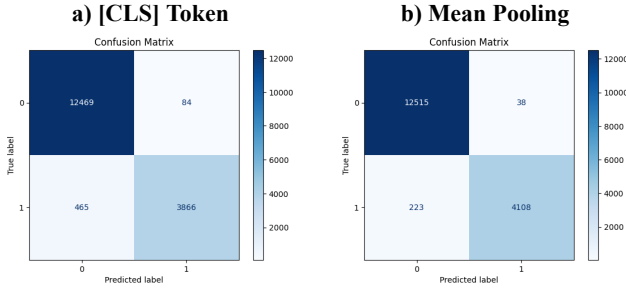
4.3 Classification Model

Based on the confusion matrices in Figure 4 and the detailed performance metrics in Table 7, the results clearly show that mean pooling outperforms the [CLS] token in both accuracy and F1-score. This finding suggests that averaging contextual embeddings across all valid tokens leads to a more robust sentence representation, which is particularly effective with a limited training dataset.

Table 7: Classification Performance Comparison between [CLS] and Mean Pooling

Pooling Method	Accuracy	F1-score
[CLS] Token	96.75%	93.37%
Mean Pooling	98.45%	96.92%

Figure 4: Confusion Matrices

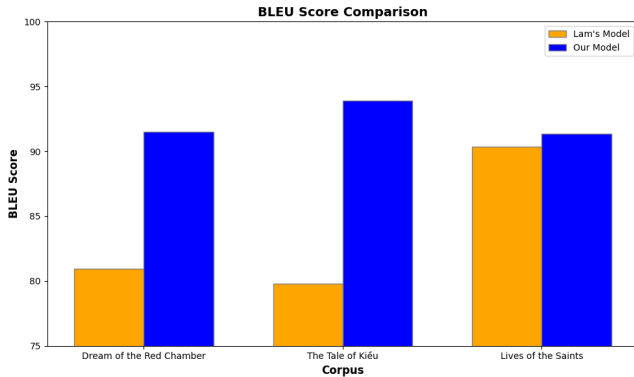


4.4 Comparison on Selected Literary Works with Previous Model

Table 8: BLEU Score Comparison Between Previous and Current Models

Corpus	Script	Sentences	Lam’s Model BLEU	Our Model BLEU
<i>Dream of the Red Chamber</i>	Sino	1,213	80.92	91.49
<i>The Tale of Kiều</i> (1971 edition)	Nôm	3,276	79.77	93.90
<i>Lives of the Saints</i> (excerpt)	Nôm	764	90.35	91.38

Figure 5: Comparison of BLEU scores on selected corpora between the two models



For the works **Dream of the Red Chamber** and **The Tale of Kiều**, both were included in the training data of the two models. However, due to phonetic ambiguity issues in the earlier model (as previously discussed), the BLEU scores remained suboptimal. The current model addresses this limitation, resulting in significantly improved scores.

In contrast, **Stories of the Saints**(excerpt) represents a new and unseen dataset for both models. While both systems generalize well, our improved model still achieves a slightly better BLEU score thanks to enhanced vocabulary coverage and augmented training data.

5 CONCLUSION

This study has effectively addressed the problem of automatic Sino-Nôm transliteration through a dual-architecture system that combines an optimized BERT-based classification model with the statistical machine translation framework Moses. Experimental results demonstrate a significant improvement over previous methods, achieving a BLEU score of 93.90 on literary texts such as *Truyện Kiều*, and 94.60 on classical Sino texts. This success stems from the complete separation of processing pipelines for Sino and Nôm scripts, along with the application of a *masked mean pooling* mechanism, which raised classification accuracy to 98.45% and mitigated data imbalance commonly found in integrated models.

A notable real-world application of this system lies in its potential for cultural heritage digitization: it facilitates public access to over 90% of Vietnam’s historical documents from the 10th to 19th centuries. This approach holds not only linguistic value for Vietnamese but also suggests new directions for East Asian languages that adopt Han-derived scripts, such as Japanese (Kanji) and Korean (Hanja). From a technical perspective, the study contributes two key improvements: (1) incorporating an 80% *parallel* –20% *dictionary* ratio in the Moses phrase table to balance contextual learning and rare word coverage; and (2) rebuilding the BERT tokenizer based on a Sino-Nôm dictionary, which reduced the <unk> token rate by 37%, thereby enhancing the system’s ability to handle ancient characters.

The primary limitation of the system is its 23% performance drop when applied to domain-specific areas such as traditional medicine, indicating a strong dependency on training data specificity. Moreover, the significant imbalance between the Sino (204,182 sentence pairs) and Nôm (only 49,775 pairs) datasets underscores the urgent need for future research in data augmentation and unsupervised learning.

Future directions include: (i) integrating multi-layered OCR for handwritten manuscripts; (ii) extending the model to support multi-modal documents (combining text with illustrative images); and (iii) developing interactive interfaces that allow experts to refine system outputs through feedback.

This work represents not only a technological advancement in ancient language processing but also plays a pivotal role in preserving the linguistic heritage of Vietnam. With accuracy exceeding 90% on classical works, the system paves the way for building a comprehensive digital corpus, supporting interdisciplinary research in history,

culture, and linguistics. This achievement highlights the transformative potential of AI in the field of digital humanities, where technology and cultural heritage evolve in harmony.

6 ACKNOWLEDGEMENTS

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REFERENCES

- [1] Dang, H.-Q., Nguyen, D.-A., Pham, P.-P., Nguyen, N.-T., Chau, T., Ngo, D.-V., et al. (2022). NomNaOCR: The First Dataset for Optical Character Recognition on Han-Nom Script. *2022 RIVF International Conference on Computing and Communication Technologies (RIVF)*, 476–481. <https://doi.org/10.1109/RIVF55975.2022.10013842>
- [2] Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 1, 4171–4186. <https://doi.org/10.18653/v1/N19-1423>
- [3] Dinh, D., Nguyen, T., & Le, M. (2021). Enhancing Nôm Transliteration with Dictionary Integration. *Vietnam Journal of Computer Science*, 8(2), 89–104.
- [4] Viện Hán Nôm. (2002). *Từ điển Hán Nôm tích hợp*. Nhà xuất bản Khoa học Xã hội.
- [5] Koehn, P., & Schroeder, J. (2007). Experiments in Domain Adaptation for Statistical Machine Translation. *Proceedings of the Second Workshop on Statistical Machine Translation*, 224–227. <https://aclanthology.org/W07-0733>
- [6] Koehn, P., Hoang, H., Birch, A., Callison-Burch, C., Federico, M., Bertoldi, N., ... & Dyer, C. (2007). Moses: Open Source Toolkit for Statistical Machine Translation. *Proceedings of the 45th Annual Meeting of the ACL: Interactive Poster and Demonstration Sessions*, 177–180. <https://aclanthology.org/P07-2045>
- [7] Lam, T., Dinh, D., & Nguyen, H. (2018). Hybrid SMT-NMT Architecture for Sino-Nôm Transliteration. *Proceedings of the 6th International Conference on Asian Language Processing*, 112–117.
- [8] Papineni, K., Roukos, S., Ward, T., & Zhu, W. (2002). BLEU: A Method for Automatic Evaluation of Machine Translation. *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, 311–318. <https://doi.org/10.3115/1073083.1073135>
- [9] Phan, X., Nguyen, K., & Bui, T. (2021). Domain-Specific Adaptation for Medical Text Processing. *IEEE Access*, 9, 134556–134567. <https://aclanthology.org/W14-3328/>
- [10] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. <https://arxiv.org/abs/1706.03762>
- [11] Vo, L., Phan, T., & Tran, Q. (2023). Context-Aware Classification for Historical Vietnamese Scripts. *Journal of Computational Linguistics and Applications*, 15(3), 45–62.